

## 4.12 Diagnostic Virology II

### ▼ Recall some viral infections that can be sexually transmitted.

- HIV (Human Immunodeficiency Virus)
- HTLV (Human T-Lymphocytic Virus)
- Hepatitis B/C
- Human Papilloma Virus
- Herpes Simplex Virus

### ▼ What is point-of-care testing?

- **Carrying out a test** using a device or test kit **in the presence of the patient** and without the need to send a sample to a laboratory.

### ▼ What may be the reasons behind a HIV test returning negative?

- The patient actually **does not have HIV**
- The **test has been performed too close to the onset of the infection**, so there are **not enough HIV antigens present yet** to be detected
- The **test has been performed too close to the onset of the infection**, so there are **not enough HIV antibodies present** to be detected

### ▼ Which further tests can be done beyond an antigen test for HIV?

- Antibody test
- PCR to amplify blood samples
- Nucleic acid tests (NAT) done to detect RNA in actual HIV.
- qrt-PCR or RT-qrt-PCR to determine viral load

### ▼ How can you test a baby for HIV infection?

- Heel prick test to obtain blood from the baby
- PCR can be done to amplify sample

### ▼ Why shouldn't we test for antibodies in babies under two years old?

- Do not test for antibodies as the antibodies can be transferred from the mother to the child during pregnancy
  - These can stay in the child's serum for up to two years
- This hence doesn't give us an accurate representation of whether or not the mother or the child has produced the antibodies in response to the pathogen.

- ▼ What further tests can be done to follow-up a positive HTLV serology test?
- **qrt-PCR** to determine **volume of HTLV** present in patient
  - This gives us an indication on **how likely the patient is to develop severe complications** as a result of the virus, and their **likelihood of transmission**